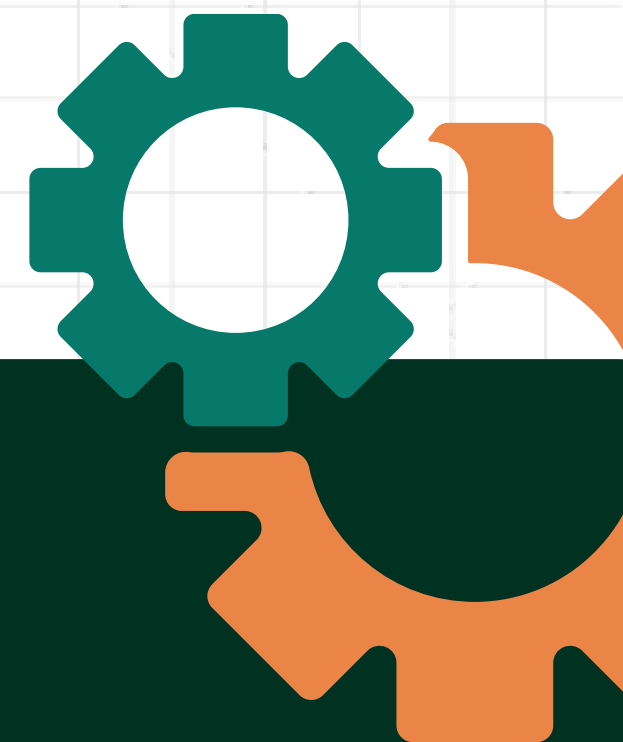
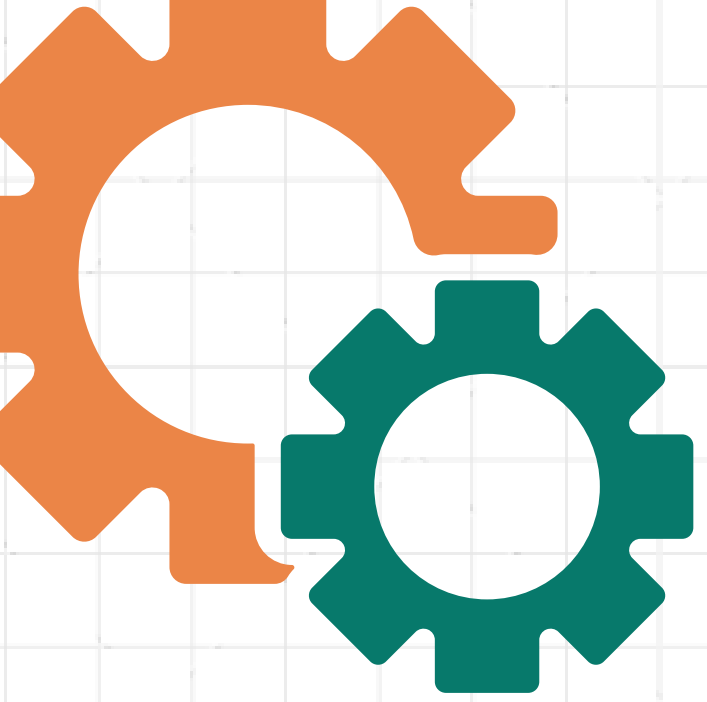




The Machine Learning Landscape

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Learning Objectives

1

Understanding
basic concepts of
Machine learning

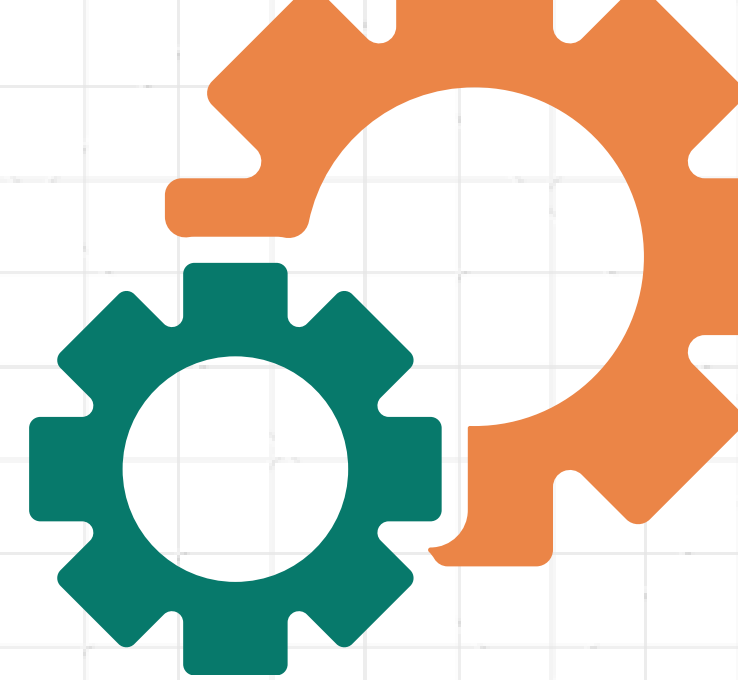
2

Get familiar with
the tools and
frameworks

3

Understanding Machine
Learning Development
Lifecycle

What is Machine Learning?



Machine Learning focuses on the use of data and algorithms to imitate the way that human learn, gradually improving its accuracy.

OR

Machine Learning used to describe a variety of tools and techniques which allow a computer program to learn and improve over time.

AI vs ML vs DL

Artificial Intelligence means creating smart machines to mimic human behavior.

Machine Learning is a subset of AI, and builds a model based on training data to make predictions.

Deep Learning is a subset of ML algorithms to solve complex problems.

AI

Reactive Machines
Limited Memory
Theory of Mind
Self awareness

ML

Supervised Learning
Unsupervised Learning
Reinforcement Learning

DL

Convolutional Neural Networks
Recurrent Neural Network
Generative Adversarial Network
Deep Belief Network

Types of Machine Learning

Machine Learning

Supervised

- Regression
- Classification

Unsupervised

- Clustering
- Dimensionality Reduction
- Anamoly Detection
- Association Rule

Semisupervised

Reinforcement

Supervised Learning Algorithms

Classification

- Decision Trees
- Support Vector Machines
- Random Forest
- K Nearest Neighbors
- Naive Bayes

Regression

- Linear Regression
- Logistic Regression
- Lasso Regression
- Ridge Regression

Unsupervised Learning Algorithms

Clustering

- K-Means
- DBSCAN
- Hierarchical clustering Analysis

Dimensionality Reduction

- Principle Component Analysis
- Kernel PCA
- t-SNE

Association

- Apriori
- Eclat

Batch vs Online Machine Learning

- **Batch Learning:**
 - Training is performed on the entire dataset at once.
 - Model is trained iteratively on the entire dataset for multiple epochs.
 - Computationally expensive and requires sufficient memory to hold the entire dataset.
 - Typically used when the dataset fits in memory and when periodic retraining is feasible.
- **Online Learning:**
 - Training is performed incrementally on individual data instances or mini-batches.
 - Model is updated continuously as new data becomes available.
 - Suitable for handling large datasets and streaming data.
 - Enables real-time learning and adaptation to changing conditions.

Instance vs Model based Machine Learning

Instance-based Learning:

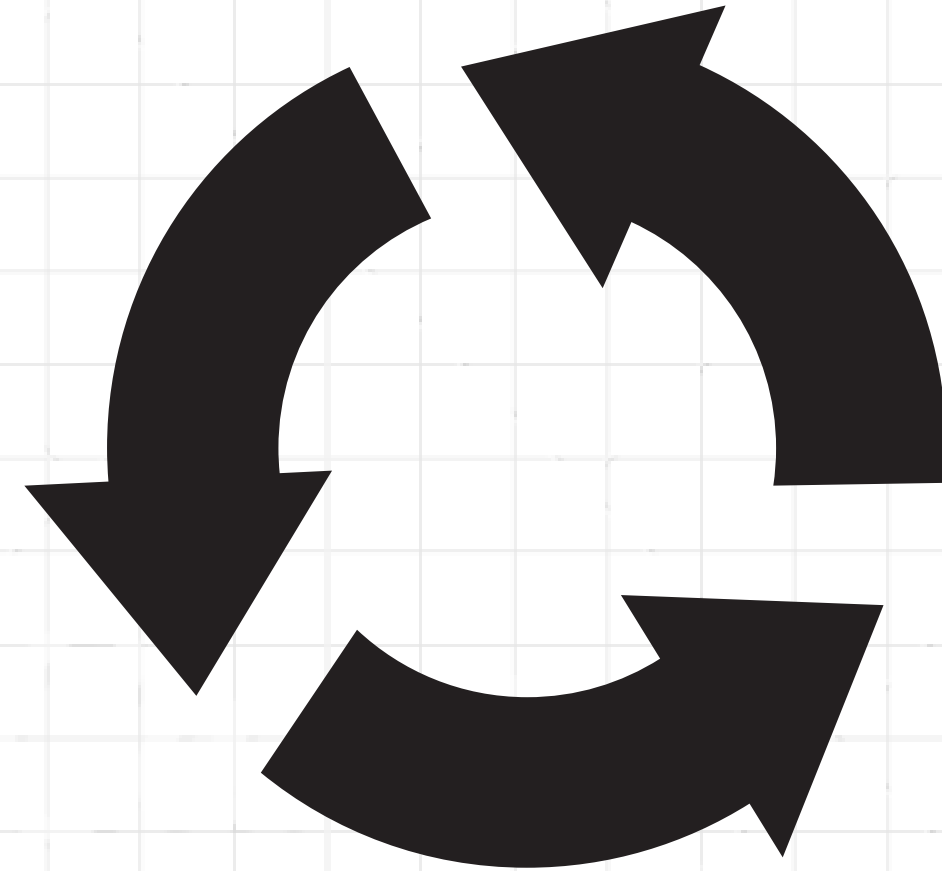
- Model makes predictions based on similarities between new instances and instances in the training dataset.
- No explicit training phase; all training data is stored and used for inference.
- Simple and easy to implement but can be computationally expensive during prediction.

Model-based Learning:

- Model learns underlying patterns and relationships from training data.
- Training phase involves estimating model parameters that optimize a predefined objective function.
- Model makes predictions based on learned parameters without directly comparing new instances to training instances.
- Typically more efficient for prediction but requires more upfront computational resources for training.

ML Life Cycle

- Frame the problem
- Gathering Data
- Data Preprocessing
- Exploratory Data Analysis EDA
- Feature Engineering / Selection
- Model Training and Evaluation
- Model Deployment
- Model Testing

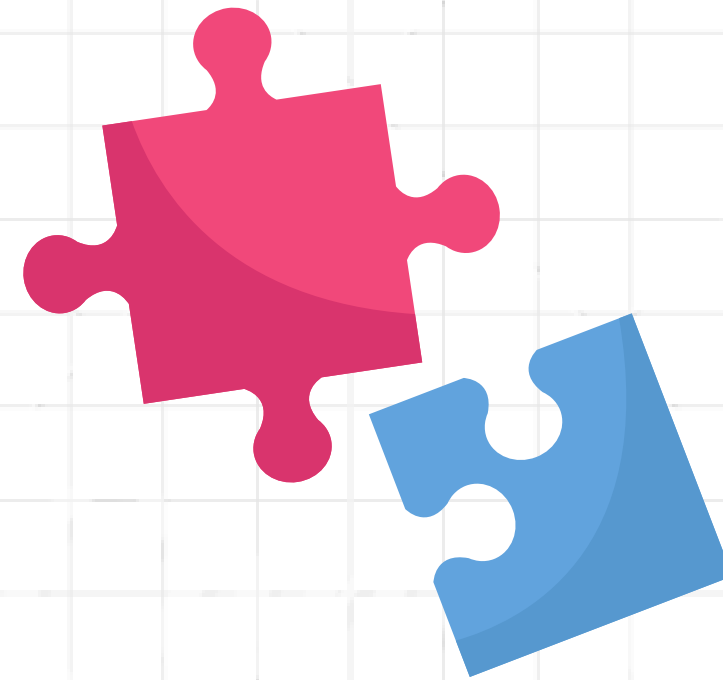


Application of ML

- Retail - Amazon/ BigBazar
- Banking and Finance
- Transport
- Manufacturing and Maintenance
- Social Media

Challenges in Machine Learning

- Data Collection
- Insufficient Data
- Non representative data
- poor quality data
- Irrelevant features
- Overfitting
- Underfitting
- Cost Involved



Thank you!